

Generalized Vibrational Analysis

Part I: Linear Algebra on a Metric Space

Goal: establish vectors, covectors, the metric, and projection operators with no reference to coordinates or physics. Raising and lowering indices arise naturally as the way the metric turns vectors into covectors.

A single space

Definition 1. Vectors and covectors. Let V be an n -dimensional real vector space with basis $\{e_i\}$. Every vector $v \in V$ has components v^i with $v = v^i e_i$, where repeated upper and lower indices are summed. A *covector* is a linear map $\alpha : V \rightarrow \mathbb{R}$, and the covectors form the *dual space* V^* . The components of a covector are $\alpha_i \equiv \alpha(e_i)$, so that $\alpha(v) = \alpha_i v^i$.

Definition 2. Metric. A *metric* on V is a symmetric, positive-definite bilinear form $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$. Its components are $G_{ij} \equiv \langle e_i, e_j \rangle = G_{ji}$, so that

$$\langle v, w \rangle = G_{ij} v^i w^j.$$

Definition 3. Lowering. The *lowering* (flat) of a vector v is the covector $v^\flat \equiv \langle v, \cdot \rangle$, i.e. $v^\flat(w) = \langle v, w \rangle$. Its components are written $v_i \equiv (v^\flat)_i$.

Proposition 1. Lowering in components.¹

$$v_i = G_{ij} v^j \quad \langle v, w \rangle = v_i w^i$$

The inner product is the contraction of a vector with the lowered form of the other.

Proposition 2. Lowering is invertible.² The map $\flat : V \rightarrow V^*$ is a linear bijection.

Definition 4. Raising. The *raising* (sharp) map $\sharp : V^* \rightarrow V$ is the inverse of $\flat$: $\alpha^\sharp$ is the unique vector with $(\alpha^\sharp)^\flat = \alpha$. Its components are written $\alpha^i \equiv (\alpha^\sharp)^i$, and the *contravariant metric* G^{ij} is defined as the component matrix of $\sharp$:

$$\alpha^i = G^{ij} \alpha_j.$$

Lemma 1. Co- and contravariant metrics.³ The co- and contravariant metrics are matrix inverses, and G^{ij} is symmetric.

$$G^{ik} G_{kj} = G_{jk} G^{ki} = \delta_j^i \quad G^{ij} = G^{ji}$$

Remark 1. Dual consistency. By construction, $(v^\flat)^\sharp = v$ and $(\alpha^\sharp)^\flat = \alpha$. The position of an index therefore tells us which object a component belongs to, and we write v_i and α^i freely.

Definition 5. Covector inner product. The inner product of covectors is $\langle \alpha, \beta \rangle \equiv \langle \alpha^\sharp, \beta^\sharp \rangle$.

¹Proof: **TODO**

²Proof: **TODO**

³Proof: **TODO**

Proposition 3. Covector inner product in components.⁴

$$\langle \alpha, \beta \rangle = G^{ij} \alpha_i \beta_j = \alpha_i \beta^i \quad \langle v^\flat, w^\flat \rangle = \langle v, w \rangle$$

Definition 6. Orthogonal projector. An orthogonal projector is a linear map $P : V \rightarrow V$, with components $(Pv)^i = P_j^i v^j$, that is idempotent and self-adjoint:

$$P^2 = P \quad \langle Pv, w \rangle = \langle v, Pw \rangle.$$

Its complement is $Q = 1 - P$.

Theorem 1. Projection onto a vector.⁵ For $v \neq 0$, the orthogonal projector onto $\text{span}\{v\}$ is

$$P_v = \frac{v \otimes v^\flat}{\langle v, v \rangle} \quad (P_v)_j^i = \frac{v^i v_j}{\langle v, v \rangle} \quad P_v w = v \frac{\langle v, w \rangle}{\langle v, v \rangle},$$

and $Q_v = 1 - P_v$ is the orthogonal projector onto the orthogonal complement of v .

Corollary 1. Projection of covectors.⁶ An orthogonal projector P acts on covectors from the right, $(\alpha P)_j \equiv \alpha_i P_j^i$ (i.e. $\alpha P = \alpha \circ P$), and this is an orthogonal projector on V^* with respect to the covector inner product, satisfying $(Pv)^\flat = v^\flat P$. In particular, for $\alpha \neq 0$, the projector onto $\text{span}\{\alpha\}$ is

$$(P_\alpha)_j^i = \frac{\alpha^i \alpha_j}{\langle \alpha, \alpha \rangle} \quad \beta P_\alpha = \alpha \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}.$$

Definition 7. Projection of a bilinear form. The projection of a bilinear form H_{ij} by an orthogonal projector P is $H(Pv, Pw)$, with components

$$(H^P)_{ij} \equiv P_i^k H_{kl} P_j^l.$$

Linear maps between two spaces

From here on, Greek indices μ, ν, ρ, σ refer to a space V of dimension n and Latin indices i, j, k, l refer to a space W of dimension $m \leq n$.

Definition 8. Linear map and pullback. A linear map $L : V \rightarrow W$ has components $(Lv)^i = L_\mu^i v^\mu$. Its pullback (transpose) $L^* : W^* \rightarrow V^*$ is $L^* \beta \equiv \beta \circ L$, with components $(L^* \beta)_\mu = \beta_i L_\mu^i$.

Proposition 4. Pullback of a surjection is injective.⁷ If L is surjective, then L^* is injective.

Definition 9. Induced metric. Given a metric on V and a surjective $L : V \rightarrow W$, the induced covector inner product on W^* is

$$\langle \beta, \gamma \rangle_W \equiv \langle L^* \beta, L^* \gamma \rangle_V.$$

The induced metric on W is the one whose raising and lowering reproduce this covector inner product.

⁴ Proof: **TODO**

⁵ Proof: **TODO**

⁶ Proof: **TODO**

⁷ Proof: **TODO**

Proposition 5. Induced metric in components.⁸ The induced contravariant metric

$$G^{ij} = L_\mu^i L_\nu^j G^{\mu\nu}$$

is symmetric and positive-definite. Its inverse G_{ij} is therefore the induced (covariant) metric on W .

Theorem 2. Minimum-norm lift.⁹ Let $L : V \rightarrow W$ be surjective and equip W with the induced metric. Define $A : W \rightarrow V$ by $A \equiv \sharp_V \circ L^* \circ \flat_W$, i.e.

$$A_i^\mu = G^{\mu\nu} L_\nu^k G_{ki}.$$

Then: (a) $LA = 1_W$, i.e. $L_\mu^i A_j^\mu = \delta_j^i$; (b) AL is the orthogonal projector on V onto $(\ker L)^\perp$; (c) Aw is the unique vector of minimum norm with $Lv = w$; (d) $\langle Aw, Aw' \rangle_V = \langle w, w' \rangle_W$.

Remark 2. Interpretation of the lift. A is simply L with both of its indices moved by the metrics. When $m = n$, $A = L^{-1}$ and $AL = 1_V$.

⁸Proof: **TODO**

⁹Proof: **TODO**

Part II: Coordinates and Derivatives

Goal: show that coordinate displacements are vectors and derivatives are covectors, so that the machinery of Part I applies to them.

Definition 10. Displacements and differentials. Let q^μ be coordinates on a neighborhood of a reference point in configuration space. Infinitesimal displacements $d\mathbf{q}$, with components dq^μ , are vectors. For a smooth function f , the differential $df(d\mathbf{q}) = \frac{\partial f}{\partial q^\mu} dq^\mu$ is linear in $d\mathbf{q}$, so df is a covector with components $\frac{\partial f}{\partial q^\mu}$.

Definition 11. Coordinate map. A second set of $m \leq n$ coordinates $q^i = q^i(q^\mu)$ has *Jacobian* and *second derivative*

$$B_\mu^i \equiv \frac{\partial q^i}{\partial q^\mu} \quad C_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial q^\mu \partial q^\nu} = C_{\nu\mu}^i.$$

We assume B_μ^i has full row rank at the reference point, so that B is surjective.

Proposition 6. Displacements push forward.¹⁰

$$dq^i = B_\mu^i dq^\mu$$

Proposition 7. Gradients pull back.¹¹ If f depends on q^μ only through q^i , then

$$\frac{\partial f}{\partial q^\mu} = \frac{\partial f}{\partial q^i} B_\mu^i,$$

i.e. df in the q^μ coordinates is the pullback B^* of df in the q^i coordinates.

Remark 3. Transformation laws. When $m = n$ and B is invertible, these become the classical contravariant and covariant transformation laws, and the contraction $\frac{\partial f}{\partial q^i} dq^i = \frac{\partial f}{\partial q^\mu} dq^\mu$ is invariant. When $m < n$, displacements can only be pushed forward and gradients can only be pulled back.

Proposition 8. Second derivatives.¹² If f depends on q^μ only through q^i , then

$$\frac{\partial^2 f}{\partial q^\mu \partial q^\nu} = \frac{\partial^2 f}{\partial q^i \partial q^j} B_\mu^i B_\nu^j + \frac{\partial f}{\partial q^k} C_{\mu\nu}^k.$$

Second derivatives therefore transform as a bilinear form only when the map is linear ($C = 0$) or at a stationary point ($\partial f / \partial q^k = 0$).

¹⁰Proof: **TODO**

¹¹Proof: **TODO**

¹²Proof: **TODO**

Part III: The Physical Model

Goal: identify the kinetic energy as the metric, and apply Part I to Cartesian, mass-weighted Cartesian, and internal coordinates.

Definition 12. Cartesian coordinates and kinetic-energy metric. For N atoms with Cartesian coordinates x^μ ($\mu = 1, \dots, 3N$), let $m_{(\mu)}$ denote the mass of the atom to which coordinate μ belongs. Cartesian displacements from a reference geometry are $q^\mu = x^\mu - x_0^\mu$. The *kinetic-energy metric* is

$$G_{\mu\nu} \equiv m_{(\mu)} \delta_{\mu\nu},$$

chosen so that $T = \frac{1}{2} \langle \dot{q}, \dot{q} \rangle$.

Proposition 9. Contravariant Cartesian metric.¹³

$$G^{\mu\nu} = m_{(\mu)}^{-1} \delta^{\mu\nu}$$

Definition 13. Mass-weighted coordinates. The mass-weighted Cartesian displacements are $\tilde{q}^\mu \equiv m_{(\mu)}^{\frac{1}{2}} q^\mu$. Mass-weighted quantities are marked with a tilde.

Proposition 10. Mass-weighted metric.¹⁴

$$\tilde{G}_{\mu\nu} = \delta_{\mu\nu} \quad \tilde{G}^{\mu\nu} = \delta^{\mu\nu}$$

Definition 14. Internal coordinates. Internal coordinates q^i ($i = 1, \dots, M$, with $M = 3N - 6$, or $3N - 5$ for linear molecules) are displacements of non-redundant functions of the geometry that are invariant to overall translation and rotation. Their mass-weighted Jacobian (B-matrix) and its derivative (C-tensor) are

$$\tilde{B}_\mu^i \equiv \frac{\partial q^i}{\partial \tilde{q}^\mu} \quad \tilde{C}_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial \tilde{q}^\mu \partial \tilde{q}^\nu},$$

and $\tilde{B}_\mu^i$ has full row rank at the reference geometry. Index-free internal-coordinate objects are marked with a prime, e.g. v' .

Theorem 3. Wilson G-matrix.¹⁵ The metric induced on internal coordinates by $\tilde{B}$ has contravariant components

$$G^{ij} = \delta^{\mu\nu} \tilde{B}_\mu^i \tilde{B}_\nu^j = \sum_\mu m_{(\mu)}^{-1} B_\mu^i B_\mu^j,$$

where $B_\mu^i = \partial q^i / \partial q^\mu$. The covariant internal-coordinate metric G_{ij} is its inverse.

Corollary 2. A-matrix.¹⁶ The A-matrix

$$\tilde{A}_i^\mu \equiv \delta^{\mu\nu} \tilde{B}_\nu^k G_{ki}$$

satisfies $\tilde{B}_\mu^i \tilde{A}_j^\mu = \delta_j^i$, and $\tilde{A}_i^\mu \tilde{B}_\nu^i$ is the orthogonal projector onto the vibrational subspace $(\ker \tilde{B})^\perp$, where $\ker \tilde{B}$ is spanned by infinitesimal translations and rotations.

¹³Proof: **TODO**

¹⁴Proof: **TODO**

¹⁵Proof: **TODO**

¹⁶Proof: **TODO**

Proposition 11. Kinetic energy in internal coordinates.¹⁷ For Cartesian velocities orthogonal to $\ker \tilde{\mathbf{B}}$ (no infinitesimal overall translation or rotation, i.e. the linearized Eckart conditions),

$$T = \frac{1}{2} G_{ij} \dot{q}^i \dot{q}^j.$$

¹⁷ Proof: **TODO**

Part IV: Vibrational Analysis

Goal: transform the gradient and Hessian into internal coordinates, set up the GF eigenproblem, and project the gradient direction out of the Hessian.

Definition 15. Gradient and Hessian. The gradient and Hessian in each coordinate system are defined by the Taylor expansion of the potential energy about the reference geometry:

$$V = V_0 + g_\mu q^\mu + \frac{1}{2} H_{\mu\nu} q^\mu q^\nu + \mathcal{O}(q^3)$$

$$V = V_0 + \tilde{g}_\mu \tilde{q}^\mu + \frac{1}{2} \tilde{H}_{\mu\nu} \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^3)$$

$$V = V_0 + g_i q^i + \frac{1}{2} H_{ij} q^i q^j + \mathcal{O}(q'^3)$$

Proposition 12. Mass-weighted gradient and Hessian.¹⁸

$$\tilde{g}_\mu = m_{(\mu)}^{-\frac{1}{2}} g_\mu \quad \tilde{H}_{\mu\nu} = m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} H_{\mu\nu}$$

Theorem 4. Internal-coordinate gradient.¹⁹

$$g_i = \tilde{g}_\mu \tilde{A}_i^\mu$$

Theorem 5. Internal-coordinate Hessian.²⁰

$$H_{ij} = \left(\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k \right) \tilde{A}_i^\mu \tilde{A}_j^\nu$$

The correction term arises because the map to internal coordinates is non-linear (Proposition 8).

Definition 16. GF matrix. The GF matrix is the internal-coordinate Hessian with its first index raised,

$$(GF)_j^i \equiv G^{ik} H_{kj},$$

a linear map on internal-coordinate displacements. Its eigenvalues are the squared harmonic frequencies ω^2 .

Proposition 13. GF and mass-weighted Cartesian eigenvalues.²¹ Let $\tilde{P}_\nu^\mu = \tilde{A}_i^\mu \tilde{B}_\nu^i$. The eigenvalues of $G^{ik} H_{kj}$ are the eigenvalues of $\tilde{P}_\mu^\rho \left(\tilde{H}_{\rho\sigma} - g_k \tilde{C}_{\rho\sigma}^k \right) \tilde{P}_\nu^\sigma$ restricted to the vibrational subspace. At a stationary point, these are the eigenvalues of the projected mass-weighted Hessian.

Definition 17. Gradient-direction projector. For $g_i \neq 0$, with $g^i = G^{ij} g_j$ and $\langle g', g' \rangle = G^{ij} g_i g_j$, the gradient-direction projector and its complement are

$$(P_g)_j^i = \frac{g^i g_j}{\langle g', g' \rangle} \quad Q_g = 1 - P_g.$$

¹⁸Proof: **TODO**

¹⁹Proof: **TODO**

²⁰Proof: **TODO**

²¹Proof: **TODO**

Theorem 6. Projected Hessian.²² *The projected Hessian*

$$H_{ij}^Q = (Q_g)_i^k H_{kl} (Q_g)_j^l$$

gives a GF matrix $G^{ik} H_{kj}^Q$ with eigenvector g^i of eigenvalue 0. Every eigenvector v^i with non-zero eigenvalue satisfies $g_i v^i = 0$, and these $M - 1$ eigenvalues are the generalized frequencies ω^2 for motion orthogonal to the gradient.

Remark 4. Comparison with Cartesian projection. *To investigate: the relation to Cartesian gradient projection (Miller, Handy, and Adams). Note that $\tilde{A}_i^\mu g^i = \tilde{g}^\mu$, so the gradient directions agree. By Proposition 13, however, the internal-coordinate result carries the curvature term $g_k \tilde{C}_{\mu\nu}^k$, so projected frequencies away from stationary points depend on the choice of coordinates.*

²²Proof: **TODO**